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Degrowth in Practice: Developing an Ecological Habitus within Permaculture Entrepreneurship

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Abstract: The literature on degrowth has suffered from only engaging with normative ideas. More recently the degrowth debate has started moving from a normative perspective to close the wide gap that has existed between normative ideas and is analysing how to link ideas to the institutional and cultural environment that shapes practices. To address this challenge, we draw on the work of Pierre Bourdieu in order to examine transformations in the habitus and forms of capital of those who decide to move into sustainable entrepreneurship through permaculture in Brazil. Permaculture represents a vibrant alternative to industrial food production and addresses fundamental contemporary social problems, such as increasing inequalities, climate change and the loss of biodiversity. The article explores the challenges faced and respective responses of those who decide to change their relationships with nature and society by becoming permaculture entrepreneurs (PEs). The paper shows that, when entering the permaculture universe, individuals start critically examining their values and ways of living, which leads to a disengagement from dominant patterns of behaviour and social expectations in order to pursue sustainable lifestyles and thereby develop an ecological habitus.

Keywords: permaculture; degrowth; Pierre Bourdieu; ecological habitus; forms of capital; Brazil



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1. Introduction

All over the world, societies are organized around the ‘growth’ paradigm [1,2], which, it is largely accepted, is independent of the level of national economic development. The growth paradigm circumvents the public arenas where different possibilities of socio-ecological orders are discussed [3]. Economic and sociological theories treat nature as the supplier of free resources. Although the capitalist system could not exist without these “free resources”, they are invisible in hegemonic discourses and economic policies [4]. Not surprisingly, the consequences of this blind pursuit of growth have included ecological and economic crises which have become the everyday condition of our lives. For example, every day, the inhabitants of big cities face the problems of pollution, violence and psychological stress.

Currently, a body of literature is emerging that deals with degrowth economics by building up a critique of the assumption in economic thought and policy-making that economic growth is vital, an absolute condition for the survival of the global political and economic system. In the field of ecological economics, new ways of understanding the links between the economy and ecology are being proposed, new ways of structuring our economy that will have far-reaching implications for the whole organization of our society: health, work, leisure, parental leave, the distribution of wealth, sustainability, corporate governance and food production. Nonetheless the degrowth literature still needs to analyze the real possibilities for transformation and the nature of the changes required [5]. The notion of ‘permaculture’ is closely related to the degrowth movement, as it is concerned with several aspects surrounding food production, defending the incorporation of different sustainable dimensions: ecological, social, economic, cultural, political and ethical.

Research has shown that these transformations are likely to have strong consequences at the individual and group levels. By exploring the becoming of permaculture en-

trepreneurs (PEs), this paper contributes to the emerging stream of research that is investigating the diversity of—and challenges confronting—sustainable entrepreneurial practice [6,7]. It also contributes to a better understanding of how degrowth takes place in the Global South by discussing the process of becoming a sustainable entrepreneur in Brazil. We connect the pursuit of a sustainable society with the internalization of environmental and ecological dispositions, i.e., sustainability requiring the development of an ecological habitus [8]. We place our research in Brazil as it is a large country, with a tradition of producing food in a very industrial way that has severe negative environmental impacts; moreover, it is also a highly urbanized society. Brazil's industrial food-production regime is a major contributor to large-scale deforestation, water and energy scarcity, increased vulnerability to climate change, concentration of land, and severe social problems [9]. This system puts unprecedented pressure on the Brazilian environment, ecosystems and society. These conditions make the investigation of permaculture entrepreneurs in Brazil quite fascinating, as these persons need to overcome several structural barriers in order to become sustainable entrepreneurs.

The debate on degrowth has only recently started moving from a normative perspective to close the wide gap that has existed between these normative ideas and analyses of how to link ideas to the institutional and cultural environment that shapes practices [10]. Several transformations are proposed by the degrowth literature, a central one being the transformation of food production. Substantial food-system transformation has become a *sine qua non* for degrowth [11–13]. In most countries and societies, we now face the challenges of feeding the population fairly and of regenerating the environment in order to enhance resilience, necessitating new pathways towards substantial social innovation, experimentation and transformation [14,15].

Current industrial food-production systems are well established and supported by a number of highly institutionalized practices, including consumer behaviour and business patterns. In different countries, a whole web of institutions and cultural norms reinforces local patterns of production and, therefore, immense institutional and cultural changes need to take place to our societies, moving them towards a new food-production landscape. Consequently, new and renewed ways of organizing agriculture are receiving increased attention worldwide in the context of sustainable food production. One example of this transformation is the agroecology and permaculture movement. Permaculture promotes a radical rupture with present modes of food production and patterns of household consumption. In this article we analyse the case of permaculture in Brazil as a form of degrowth.

Permaculture represents not only a radical alternative for food production but also a radical transformation in social values [16] and promotes new imaginaries for how to deal with environmental problems [17–19]. Given the increasing popularity and ecological importance of permaculture [20,21] it deserves sociological attention, as it is increasingly attracting people from many different social and cultural backgrounds, and not primarily from farming. The relevance of permaculture to the food transition draws attention to the issue of world views [22] and demands further investigation of the social aspects of permaculture [23–25]. Permaculture is a paradigm case of a social practice that envisages a rupture with established patterns of production and consumption, as well as demanding radical lifestyle changes from those who intend to adopt its practices [9,26]. As such, it becomes an ideal social practice for studying the potential changes required for an ecological transformation [10,27].

We started asking how permaculture practitioners construct the process of pursuing a sustainable lifestyle, and how they use their previous forms of cultural and social capital and acquire new ones? This article shows that permaculture is a tangible, applied practice for sustainability. This paper connects ideas about forms of capital, habitus, and the choices and lifeways of permaculture entrepreneurs (PEs). The article shows how PEs struggle to create alternative food systems which are linked and able to support new ways of organizing their life choices and consumer patterns. Our analysis shows that potential PEs need to abandon most of their previously accumulated forms of capital. This makes

the transition a highly risky venture, as they need to accumulate new forms of capital and profoundly modify their habitus. This process seems to be complicated due to the fragile characteristics of the Brazilian welfare system in the areas of education and health.

The remainder of this paper is structured as follows: We begin by reviewing the literature on the concepts of habitus and forms of capital, in line with our own theoretical perspective. This is followed by an account of the research design, an introduction to the concept of permaculture, and a description of the local conditions of permaculture in Brazil. We describe the recent evolution of permaculture in Brazil and present our results, divided into three subsections. The first sub-section describes the reconstruction of habitus and the second the ecological habitus and capitalist relations. The intention here is to show the transformation of the habitus of those middle-class Brazilians who adopt permaculture and to show the increasing resistance of permaculture practitioners to capitalist relations of production. The third section deals with permaculture entrepreneurship and welfare institutions. The welfare institutions are shown to limit the choices of some families when moving into permaculture. Private education and health insurance are important services for the Brazilian middle classes, we show that, when moving into permaculture, the income level becomes more insecure but new ways of understanding and dealing with consumption also emerge with an ecological habitus. The discussion section analyses our findings and discusses the contribution of the article, its limitations and future research directions.

2. The Emergence of an Ecological Habitus

To undertake a thorough review of Bourdieu's work would take us beyond our aims here [28]. Our objective in this section is rather to provide a brief overview of his central theoretical concepts of habitus and social and cultural capital. The concept of habitus was originally articulated by Mauss, but developed subsequently and popularized by Bourdieu [29]. For the latter, habitus is a set of dispositions that inclines agents to perceive, analyze and react in certain ways; it provides a basis for perception, appreciation and action. It shapes the tendencies to interact with other actors and social structures in particular ways [30]. Habitus and social structures dynamically affect and create each other in such a way that dispositions and social positions are mutually congruent. Individuals are more prone to reproduce the social structures that support their habitus. According to Bourdieu: "the habitus is necessity internalized and converted into a disposition that generates meaningful practices and meaning-giving perceptions. That is why an agent's whole set of practices (or those of a whole set of agents produced by similar conditions) are both systematic, inasmuch as they are the product of the application of identical (or interchangeable) schemes, and systematically distinct from the practices constituting another life-style" [31].

The field of permaculture, like many other fields, requires and supports a very particular form of habitus, which carries its particular world view. The field asks for the particularities of a habitus which can play up to its expectations and has the necessary skills and knowledge (discussed below as cultural capital). The habitus as a world view is composed of dispositions and practices that produce and reproduce social structures in the field of action. It integrates the individual into the field, but it may also generate the lack of integration of the disparate experiences that make up a biography. In certain instances, the habitus can be built on contradictions, tensions and even instability. Tensions and instability become the drivers of social criticism, which might generate a new world view demanding social change. As Reay asserts:

"The range of possibilities inscribed in a habitus can be envisaged as a continuum. At one end, habitus can be replicated through encountering a field that reproduces its dispositions. At the other end of the continuum, habitus can be transformed through a process that either raises or lowers an individual's expectations. Implicit in the concept is the possibility of a social trajectory that enables conditions of living that are very different from initial ones." [32]

For us, it is important to stress that, particularly in moments of a lack of integration with their fields of action, individuals might suffer from internal conflicts and powerful emotions leading them to question taken-for-granted assumptions. For example, this lack of integration with contemporary consumer societies may be generating the drivers for changes in world views by stimulating alternative views of social and ecological problems. Bell calls this moment a phenomenological break or phenomenological rupture that compels or enables actors to break free from dominant social structures and culture [33], when actors may realize a new world view is possible and starting enacting it. Entering new social movements such as permaculture might trigger the development of a social critique of consumerism and capitalist relations of production and nurture the formation of a renewed habitus, based on an ecological world view concerning the relationship between humanity and nature.

New world views come with renewed dispositions, practices and consequently actions in the struggle to change social structures. But before gaining new dispositions, when taking up the permaculture movement, actors may face a lack of integration with their present world views, practices and dispositions. They may be criticized, or even excluded from their current networks. The dispositions that shape people's ecologically relevant tendencies, sensitivities and, consequently, choices and practices insofar as these are interrelated with material conditions. Here the ecological habitus characterizes a world view which takes into consideration the environmental impacts of the individual's choices and that leads the individual to see beyond the consumerist view. Thus, the ecological habitus refers to pro-environmental practices and ways of being, not only what to consume, but how to produce, how to live and what to value as acceptable. Ecological habitus is more than green consumption and, in the case of permaculture, it represents a transformation of the house from a place of consumption to a place of production. Consequently, the permaculture movement as a form of praxis (a unity of theory and practice that engages critical reflection) becomes a nurturing field for the ecological habitus.

To complement our theoretical concepts, below we discuss different forms of capital as a preliminary to analysing the different facets of the transformation that takes place when individuals take up permaculture. The different forms of capital are important in understanding the changes that individuals need to make in order to enter and establish themselves in the field of permaculture, and the transformation in the previous forms of capital and the creation of new ones as conditions for the transformation towards an ecological habitus.

3. Building New Forms of Capital to Support the New Ecological Habitus

The analysis of different types of capital is central to Bourdieu: social capital, cultural capital, symbolic capital, political capital and economic capital, to mention but the most significant variants [34]. In "Forms of Capital", Bourdieu develops a notion of capital that, besides material exchanges, also includes "immaterial" forms of capital (intangible forms) and, more specifically, cultural and symbolic capital. Bourdieu discusses how the different types of capital are acquired, exchanged and converted into other forms. The structure and distribution of capital also represents the inherent structure of the social world. It is thus necessary to reintroduce capital in all its forms and not solely in the one form recognized by economic theory. It is also necessary to understand how individuals can acquire new forms of capital when moving from one field to a new one, as is the case in our investigation when urban middle-class individuals move into permaculture.

For Bourdieu, cultural capital has three states: embodied in the individual (as a type of habitus), objectified in cultural goods, or institutionalized as academic credentials or diplomas [35]. Schools function not only as sites of the distribution of cultural capital, they also represent sets of valorisation and legitimization of the cultural capital of the different classes. Cultural capital can consist of informal academic standards, which are also class attributes of the different classes. Cultural capital is an indicator and basis for class position, the attributes of which, namely behaviour and preferences, are mobilized

for social selection [36]. Cultural capital thus represents what the individual knows and is able to do, and can be increased by investing time in self-improvement in the form of learning [37]. Consequently, we need to investigate how the move into permaculture affects the amount of cultural capital that individuals previously have and how they are able to translate their previous forms of capital into new ones. Cultural capital requires dedication and time for its accumulation and is not easily lost. Thus, permaculture entrants must invest time and resources to build new cultural capital. Over time, cultural capital becomes a part of habitus and cannot easily be transmitted, though it can easily be read by other individuals. However, moving into permaculture means a large move into a new field, one with different values, and asks for unique forms of cultural capital and consequently a modified habitus. Permaculture entrepreneurship is likely to require new forms of capital, therefore we need to inquire how these new forms of cultural capital are obtained and if and how PEs translate their previous forms of cultural capital into new ones and use them.

Social capital consists of all actual or potential resources linked to a network of more or less institutionalized relationships of mutual familiarity and identification. It refers to the possibilities and resources that an actor can use when embedded in a network of social relations. Bourdieu defined social capital as the aggregate of the actual or potential resources that are linked to the possession of a durable network of more or less institutionalized relationships of mutual acquaintance and recognition. Social networks are not durable and need to be fostered over time, while networks are not necessarily flexible with regard to exit and return, as an exit may imply that an actor cannot be reconnected to the network. In spite of being possessed by individuals like private property, the diverse forms of capital are embedded in their context and do not necessarily present the same value outside its field.

Actors adopting permaculture are likely to have their social capital reduced and to find it necessary to build new social relations and a sense of belonging to the permaculture community. Actors need to develop strategic improvisations in order to respond to the opportunities and constraints opened up by different situations as when entering the permaculture movement. The question then becomes, what is made of the habitus and different forms of capital of those who move into new social constellations, new conditions that are radically different from those they are used to, as when entering the new field of permaculture and building up their ecological habitus?

The argument proposed here is that actors are able to re-invent their habitus, thereby creating a new world view in order to scrutinize opportunities and constraints from a different perspective. Because of this change in world view, they will be able to choose and construct a new set of possibilities that they want to pursue and the constraints they want to overcome. When reconstructing their habitus, previously inconceivable possibilities are revaluated, the change in habitus creates a new range of choices, and as new opportunities emerge, many previous options simultaneously become unavailable and even offensive.

Permaculture practitioners use a period of transition to radically adjust their previous cultural maps, reimagining the world they want to live in. When embedded in the new community, working with and living from permaculture, their habitus develops a lack of integration between their present and past experiences, paving the way to new and unexpected biographies. The discomfort, triggered by the encounter with permaculture values, drives actors towards an epiphany, a phenomenological break. Eventually these individuals seek to transform their lives, rejecting and struggling to transform their current forms of accumulated capital and eventually their habitus. Newcomers to permaculture faced great strains as they moved away from their cultural and social capital, which, at that time, were tied to their formal careers in the labour market. Our analysis indicates that their progressive attachment to permaculture happens simultaneously with the transformation of their habitus, which, in general, is a very difficult and arduous life change receiving weak support from their previous social networks. It is achieved by entering new communities and building strong social ties to the permaculture community, which supports these radical life changes.

4. Research Design and Methodology

The research approach was case-study-based. The examples here involve an in-depth case study of permaculture practitioners, entrants and permaculture entrepreneurs during a period of three years. The paper describes the experience and determines the extent to which existing theories help us understand such cases or require modification [38]. Our intention was to develop a processual and holistic approach [39]. Our understanding of permaculture is informed by the social constructivist stream in ecological economics (EE) [40–42]. Accordingly, we attempt to embed the operation of economic activities in natural science perspectives, as well as socio-political processes [43,44] that recognize that human activities are bounded and produced by complex social and biophysical relations, thereby emphasizing the reduction in economic activity to its social and ecological dimensions. Our methodology is based on a case study that uses hermeneutical approaches for an in-depth understanding of the processes of social and agricultural change within permaculture.

In order to analyse the cultural transformations of permaculture entrepreneurs, we use hermeneutic or interpretive inquiry, a qualitative research method that focuses on social and discursive practices and processes of interpretation. Hermeneutics is the philosophy and practice of interpretation [45–47]. In this tradition, the situatedness of human action in relation to the broader society produces meaning [48]. The article draws on Bourdieu's concepts of habitus and capital in order to inquire into the interactions that occur between actors in an effort to interpret and understand the trajectories that lead them into permaculture. When inquiring and analysing the transition to becoming a PE, conceptualizing the situatedness of human action implies identifying a network of actors and their relations, and the emergence of new forms of capital among them. Our research was not designed to include a set number of participants; rather, as Moules claims, hermeneutic research is not validated by numbers, but by the comprehensiveness of examining the topic under study and the richness and depth to which the interpretation spreads understanding [46].

5. Data Collection

The data were collected during three fieldwork trips undertaken by the author in the state of Ceará, Brazil, starting in 2018 and ending in 2021. We consider the study to be a longitudinal field study because seventeen entrepreneurs were interviewed at different moments during this time. Table 1 gives a list of the interviewees. It was arranged to conduct these interviews to take place once every year, the intention being to discuss the evolution of the interlocutors' experience with permaculture. We used several different sources in collecting data:

- Participation in an annual permaculture convergence, the annual conference of permaculture practitioners in the region. Over fifty permaculturalists gathered for three days in a permaculture homestead to exchange information, discuss issues important to the permaculture movement, display products, hold several workshops on how to create and improve resilience, and exchange seeds. Among the important issues discussed were permaculture education and certification, hosting for students and banking seeds.
- Extensive field notes were collected of the conversations held during the three-day convergence with permaculture entrepreneurs and permaculture students.
- Video recordings were made of presentations during the convergence.
- Three-hour visits to four permaculture homesteads.
- Interviews with permaculture entrepreneurs outside their homesteads.
- Participation in public seminars and workshops.
- Extensive use of interaction over social media (WhatsApp and Instagram) with PEs over two years.
- Archival materials (educational material, newspapers, YouTube videos, Instagram, and permaculture magazine articles and newsletters).

Table 1. List of interviewees.

Interviewee's Initial Codes	Gender	Status	Activities
HT	Male	Owner	Organic Food Production, Educator
CL	Female	Freelancer	Art Producer
PT	Female	Freelancer	Art Producer
MR	Male	Co-Owner	Organic Food Production, Education
OF	Male	Owner	Organic Food Production, Hospitality
MP	Female	Freelancer	Consultancy
MA	Female	Owner	Consultancy and Education
PE	Male	Owner	Ecologist, Consultancy
ME	Male	Owner	Organic Food Production
PO	Female	Co-owner	Consultancy
PW	Male	Owner	Organic Food Production, Education
TR	Female	Freelancer	Biologist, Education
CD	Female	Co-Owner	Organic Cosmetics Production, Education
SM	Male	Owner	Organic Food Production
AR	Female	Owner	Organic Cosmetics, Education
AT	Male	Owner	Organic Food Production, Consultancy
CAM	Female	Owner	Hospitality

The twelve hours of recorded interviews were transcribed; interviews with permaculture convergence participants were organized as field notes. The visits to permaculture homesteads focusing on entrepreneurs' trajectory from urban middle-class individuals to permaculture entrepreneurs and their current work and skill development took, on average, three hours. The resulting material was presented and discussed with those involved, and their feedback was incorporated into the current version.

6. Data Analysis

We employ a thematic analysis of narratives in order to examine the relationships between the diversity of actors and their experiences with permaculture. In coding we identified seven codes expressing the evolution of permaculture entrepreneurship: employment security, uncertainty, conflicting values, immersion in the permaculture universe, transitioning lifestyles, acquiring skills, and having an ecological habitus. These codes were then synthesised into four second-order codes, which were inspired by the current literature and our previous research: sources of change, the emergence of conflicting expectations, creating new capital and cognitive change (Table 2). For example, we combined statements related to "employment security" and "uncertainty" with "the emergence of conflicting expectations".

Table 2. Thematic analysis of narratives.

First Order	Second Order (Theoretical Concepts)	Examples
Employment security		"I had a very well-paid job, a secure career, but I would lie awake in the dark weighing up whether to move away from my job and apartment" MA
Uncertainty	Emergence of conflicting expectations (different fields of action)	"The good life has been sold as travelling to paradisaical places, a big house, luxury, but in order to get all those things you need to sell your life, you need to give away what is most precious, your time, your freedom." "People ask me—Oh, aren't you going to look for a job?—and I said no." CL

Table 2. Cont.

First Order	Second Order (Theoretical Concepts)	Examples
Conflicting values	Sources of change (the transitions between fields of action)	“People talk a lot about how bad things are, but no one seemed to do anything to change them, so when I started reading about agroecology, permaculture . . . I started asking myself where I belonged in the world, asked questions about the wider structures that dictate how I lived my life” MA “In the beginning of my journey, when starting closing doors and throwing away the keys, I was beaten on all sides by the guardians of the imposed world; it would have been impossible without poetry.” HT
Acquiring skills	Creating new capital (the development of social and cultural capital)	“Permaculture for me is a new journey, I believe for most of us; there is no such a thing as a definitive answer. We are experimenting, so it’s important to have peer support, other people like myself” PW
Transitioning lifestyles		“We have totally different backgrounds. I worked in marketing, and my wife was a nurse, [but] we have made a transition to permaculture. It took some years: we moved more and more into permaculture and at the same time moved gradually away from our previous careers.” MR
Immersion in the permaculture universe	Cognitive change—The emergence of an ecological habitus.	“Some books have been quite influential in my life. Small is Beautiful has been a very important one, but it is also a question of timing . . . ” AT
Ecological habitus		“Permaculture changes something inside. Because the question of permaculture is like a new perspective on life . . . the transformation is internal” CL

7. Permaculture

Permaculture was born out of a combination of a humanist tradition with the environmentalist and pacifist movements of the late 1960s. The term ‘permaculture’ is a combination of the words ‘permanent’ and ‘agriculture’, and refers to an integrated system of eco-environmental design linked to the ideal of a predominantly perennial form of agriculture. It was defined by one of its co-creators, David Holmgren, as “Consciously designed landscapes which mimic the patterns and relationships found in nature, while yielding an abundance of food, fibre and energy for [the] provision of local needs” [49]. Fadaee claims that the degrowth movement could, to some extent, be understood as a spin-off from the permaculture movement [50]. The significance of the permaculture movement is discussed as a solution-oriented movement in the face of our global problems [18]. Permaculture therefore originated as a methodology governed by principles derived from meticulous observation of nature to the conscious design of social and social-ecological systems [51]. It pursues the development of habitats that mimic the ways natural systems self-organize for productivity, resilience and adaptability [52–54]. Permaculture refers not only to a food-production system based on diversified farming systems, but also to a world view and ethical attitude that emphasises sustainability (see Table 3). The United Nations Convention to Combat Desertification (UNCCD) considers permaculture to be one of the main agricultural activities preventing land degradation [55]. The European Commission

calls permaculture “one of several prototypes of sustainable agro-ecology systems” in its Horizon 2020 program. Permaculture is a holistic sustainability concept based on a practical design process consisting of twelve principles [52], which make use of technical objects (technologies, materials and management and construction techniques) that contextualize its development and reject the homogenization of solutions and spaces. Permaculturalists work to increase energy efficiency, reduce waste production, fight generalized alienation, and avoid the overexploitation of living beings and nature. In sum, permaculture strongly discards most of the characteristics that have marked the development of the industrial society and its complementary forms of economic organization [23,56].

Table 3. Permaculture Ethics.

PC Ethics	
Care for the Earth	To promote a life in harmony with Planet Earth by: (1) maintaining a living soil; (2) stewardship of the earth; (3) supporting biodiversity; (4) defending the sacredness of life.
Care for People	Permaculture pursues the benefits of human beings—not merely for nature conservation.
Fair Share	Set limits to consumption and reproduction, and redistribute surpluses. The pursuit of long-term social justice. Every living being (not only human beings) have the right to enjoy their integrity. Alternative economic maxims such as the economy for the common good, ecology and social issues belong together.

Permaculture can be characterized approximately as being composed of four interrelated characteristics [51]. First, it may be seen as a world view that defends and constructs the potential of synergistic effects on human well-being and ecosystem health, and is also composed of moral and ethical elements [19]. Second, it is a designed system based on ecological and systems-thinking principles, and recommends specific tools for analysing and selecting appropriate practices that are sensitive to the respective ecological site conditions and goals for its utilization (see Table 4). Third, permaculture is based on interdisciplinary knowledge and techniques, as such being proposed as a framework of best practices for modelling ecological systems that aim at optimizing relationships to increase system health and mutual benefits among system participants and elements (both living and non-living) towards rebuilding resilient local communities [51,52,57]. Fourth, permaculture is supported by and grows within self-organized networks and other bottom-up movements cooperating across local, regional and transnational boundaries [51].

Table 4. Holmgren’s 12 permaculture principles.

Holmgren’s 12 Permaculture Principles		Short Description
1.	Observe and interact	Before starting on the landscape design, observe what had happened previously on the site, and interact with people who had been working and living there previously. Get to know the site before starting to modify it.
2.	Catch and store energy	Minimize the use of imported resources. Design the landscape to maximize the capture of water, sunlight, soil, and biomass in order to become more resilient.
3.	Obtain a yield	Efforts must create value. Design systems that avoid wasting energy and resources, being able to obtain a yield in a sustainable way.

Table 4. Cont.

Holmgren's 12 Permaculture Principles		Short Description
4.	Apply self-regulation and accept feedback	Accepting feedback is fundamental to assessing the effects of interventions and improving the design. Feedback is critical to learning the earth's limits.
5.	Use renewable resources and services	Use the power of the sun, the wind and other renewable sources to provide energy, grow food and regenerate the environment.
6.	Produce no waste	Reuse first and recycle all possible materials. The system must be designed to avoid wasting effort, thus increasing its efficiency. Re-purpose as much as possible. The generation of waste has negative impacts in several activities.
7.	Design from pattern to detail	Observe patterns in nature, and leverage the observed patterns. Study the pattern of what sustainable living might be and then refine into the detail appropriate for each particular site.
8.	Integrate rather than segregate	Developing good relationships with other people and organizations in the area. Good relationships among people support a more peaceful, equitable society.
9.	Use small and slow solutions	Make essential needs more local. Simplify technological alternatives. Become less dependent on complex solutions and imports.
10.	Use and value diversity	Biodiversity supports healthy ecosystems. Try to diversify crops and energy sources. A diversity of people is also central to an equitable society.
11.	Use edges and value the marginal	The edges are often overlooked. However, the interface between things is where the most interesting events take place and are often the most valuable, diverse and productive elements in the system. Pay due attention to the diversity found in the margins.
12.	Creatively use and respond to change	To be aware of the changing landscape and conditions, as well as of the resources available in order to be resilient in responding to these evolving changes.

The permaculture entrepreneur (PE) is defined here as the individual who produces and sells products and services, produced and created using permaculture principles at the market place. We focus on individuals who live in permaculture homesteads producing and selling agricultural inputs, ornamental plants, organic cosmetics, vegetables and food products. They also provide services locally, such as holding permaculture workshops, teaching permaculture principles and techniques, hospitality, ecotherapy, design and consulting, and construction. These individuals are established as permaculture producers or are committed to using permaculture to generate their main source of income. We also interviewed permaculture entrants, those who have the intention to move into permaculture but are not yet established as such.

Introduction to the Case Study: Permaculture in Brazil

When describing Brazil and, in particular, its northeast, where our study is located, it is necessary to emphasize that it is a very poor region, one with profound social and economic inequalities. In recent decades local state agricultural policies have supported the rapid and powerful development of agribusiness. The state of Ceará, where our research was located, has become an important producer of a diverse range of agricultural products produced on a large scale using industrial technologies and agro-industrial strategies: shrimp, fish, flowers and fruits. This model suits the concentration of land in the hands of a few owners. Unlike the main agro-industrial producers and small traditional farmers, PEs in Brazil are, in general, middle-class individuals struggling to save money to buy a small piece of land relatively close to urban centres in order to start producing food and live in an ecologically

sustainable fashion. PEs have formed self-organized groups of permaculture producers, with strong levels of cooperation, support activities and exchanges of knowledge and resources. These entrepreneurs produce a variety of products for the local market, such as kombucha, ecological cosmetics, ecological fertilizers, ecological bread, fruits and legumes. Their income is a combination of products sold and services provided locally. Among the services are teaching permaculture techniques, teaching gardening, workshops on waste reduction, and online workshops on how to produce permaculture products.

Permaculture has existed in Brazil since the 1980s, but due to the concentration of land in Brazil, permaculture principles (techniques and practices) have been restricted to a small group of urban middle-class individuals for whom the change in lifestyle and the construction of self-managed and sustainable spaces represent a strategy pursued against the situation of alienation and generalized estrangement experienced in capitalist modernity. Below we present the experiences of individuals who, after taking up permaculture, have decided to leave behind both their urban lifestyles and their previous professional careers. The family income is generated from the earnings from selling permaculture products and services and is supplemented by other activities and incomes not necessarily connected to permaculture, but in several cases the main source of income is their own products and services.

8. Results

8.1. The Transition: Constructing an Ecological Habitus and Acquiring New Forms of Capital

“In the beginning of my journey, when starting closing doors and throwing away the keys, I was beaten on all sides by the guardians of the imposed world; it would have been impossible without poetry.” (HT)

Individuals interested in permaculture need to reinvent their lives, habitus and accumulated forms of capital in order to become permaculture entrepreneurs. It is a long and difficult journey, and there are many barriers: economic, social and ideological. It is a learning journey, as they need to acquire new knowledge and new skills to construct a new reservoir of cultural capital. These knowledge and skills are not easily found in the market place. These individuals also need to transform the way they understand and value their material achievement and ambitions. At the beginning of their journey, they start asking unusual questions about the relationship between their living conditions and the way their relationship to nature is conceptualized in society. Moreover, they also start critically analysing how they themselves produce and reproduce environmental and social degradation. The criticism of the consumption society found in permaculture resonates with degrowth arguments against the impossibility of maintaining a growth economy forever. Permaculture practitioners echo the arguments of ecological economics about the ecological conditions of human economies, as permaculture supports the idea of reduction throughput, downscaling the economy towards a sustainable, just, and participatory steady-state society. In the interviews the links between degrowth and permaculture were made very clear, even though the word ‘degrowth’ was not directly used by PEs.

HT is one of the most influential characters in the local permaculture movement in Ceará. Like many other interviewees, he presents his personal life story as an emancipatory journey in which he struggled to free himself from the prison represented by the capitalist labour market. He was a typical middle-class urban professional, working as an information system analyst in a multinational corporation, when he started cultivating a garden in his house, and then also started recycling his own waste, both organic and inorganic. Then he began having doubts about his own way of life and his world view, and capitalist social relations became a barrier to a decent life for him. Feeling a strong discomfort with his life and job, he was in pain. *“I realised that I had a miserable life; I could not cope with it anymore.”* Over time this internal discomfort evolved and nurtured in him an intense desire to become a full-time agroforestry producer. There was a long way to go before becoming a permaculture entrepreneur, in fact some four years, until he sold his house and bought a homestead sixty kilometres from the city centre. He now lives and works full-time as an

agroforestry/permaculture entrepreneur. His mantra is that *“the imposed world transforms food into waste; in my world, we transform waste into food; for an imposed world it is the opposite.”* About his journey, he says:

“When I decided to stop trading life for the end of life and trying to live only on the emancipatory skills that I had cultivated with the help of many friends, the risks and difficulties were all in my cognitive maps and spreadsheets that I nurtured for a few years. Everything was foreseen. Even the day the money would run out, and I would go back to being scared and overwhelmed by it. But there is no cognitive map, spreadsheet or plan for domesticated uncertainty. In the process, I was fortunate to have many friends, among them one who is now dead, helping me to understand and deal with uncertainties. He told me that the way of life I intended to abandon was far more uncertain than the one I intended to embrace. The previous one was uncertain due to its nature (exchanging life for the end of life). The one I embraced is uncertain, not as to its nature, but only as to its possibility. Possibility that has been in effect for five years, but every day...one at a time. As in the prayer that awaits the realization of the possibility of each day: “...Give us this day our daily bread...” Is such a way of life uncertain? Yes...like life: uncertain about the possibilities for effectuation, not about nature.” HT

As HT states, the exchange of information and knowledge is central for succeeding in the transition to becoming a PE. “Emancipatory skills” is a wonderful concept for the cultural capital he needed to accumulate, as they are the instruments with which to escape from the prison that is the labour market. The cultivation of emancipatory skills takes place within a community of friends, where social and cultural capital grow together. In order to pursue a degrowth agenda as a PE in practice, to effectively reduce throughput, it is necessary to learn how to do it within permaculture. It is therefore essential to acquire cultural capital that can be used in permaculture and to gain knowledge in different areas: ecology, agriculture, recycling, ecological construction, health and the economy. This cultural capital is not easily available in schools, and these interviewees were already professionals with quite a different set of skills. The new cultural capital comes with information and knowledge, which are pursued in parallel with a permaculturalist world view. Learning how to see the relationship between human and nature from a new perspective is part of the construction of an ecological habitus. To support transformation through learning, the permaculture community organizes events, the annual convergence, websites and social media groups where people can meet, organize training trips and find a place to live while working as trainees. Permaculture entrants can spend extended periods of time on permaculture farms and experience and learn about permaculture on site. Those who cannot travel due family obligations or jobs can attend permaculture design courses and put what they have learned into practice on their own homesteads. For students of permaculture, the changes in their world views come by immersing themselves in a new social community, which is driven by ecological values and emerging alternatives to capitalism. When professionals from a high middle class and with well-paid jobs move into permaculture, they review their values and the transformation of their world views:

“People talk a lot about how bad things are, but no one seemed to do anything to change them, so when I started reading about agroecology, permaculture ... I started asking myself where I belonged in the world, asked questions about the wider structures that dictate how I lived my life. I had a very well-paid job, a secure career, but I would lie awake in the dark weighing up whether to move away from my job and apartment. I started seeing things in a totally different way ... a new car was no longer an achievement, a reason for enjoyment. It became painful to look at big expensive cars. They represent the loss of life, destruction of nature, diseases, suffering ... I began to realise that things which were good for me were likely to have a deleterious effect on nature and on society I saw that some people were trying to find answers to the current problems. Since then, I have changed a lot in my life” (MA)

“The good life has been sold as travelling to paradisaical places, a big house, luxury, but in order to get all those things you need to sell your life, you need to give away what is most precious, your time, your freedom. But most people do not realise this.” (SM)

“Some books have been quite influential in my life. Small is Beautiful has been a very important one, but it is also a question of timing, to read a book, to have a conversation, in the right moment of your life, when you may be open for a radical change, or when you need to make radical change to feel alive” (AT)

In Brazil, as in many other countries, a luxury car is a symbol of success, achievement and a prestigious position. However, individuals with a transformed habitus start seeing a luxury car as a sin against nature. Those who develop an ecological habitus rethink the role played by consumption in their lives, as well as the consequences for their lives of pursuing success in a capitalist society. The permaculture critique of capitalism and the consumption society is very close to the critique put forward by the degrowth community, and in permaculture, this critique is turned into practice. Permaculturalists battle to organize their lives according to new values, searching for different options and constructing new possibilities in permaculture, organic agriculture, organic construction, ecological food production, art and education. The ecological habitus re-evaluates the sources of success, what makes a good life and what should be avoided. They construct possibilities to move away from an urban life and a salaried job. The economic culture promoted by permaculture is very close to degrowth reasoning, close to what Carlsson and Manning label “nowtopias”, the conscious withdrawal from capitalist culture with a resolute rejection of its value form [58,59]. PEs critically address the capitalist culture, aiming to reject its value form, very similar to the degrowth discourse. One instance is the case of CL, a young single woman who decided to move away from her job in order to learn more about permaculture and start her own homestead. Her family was quite resistant to her choice, as she explains:

“People ask me—Oh, aren’t you going to look for a job?—and I said no.—So you’re not going to work anymore?—I said I will, but I’m not looking for a job. I’m going to do something of my own, because I don’t want to work for anyone anymore, because I think I’m capable of doing something of my own. And what is this something? It can be anything, anything that I want to do and that I dedicate my time, my attention, my commitment [to], I will do it, and it will happen, and I will make money out of it” (CL)

CL had tried to learn permaculture by living in permaculture homesteads and travelling in Brazil doing volunteer work in permaculture homesteads for six months, as a result of which, she came to understand permaculture as a radical life-changing perspective:

“Permaculture changes something inside. Because the question of permaculture is like a new perspective on life . . . the transformation is internal, and I found my way through permaculture and I saw it . . . it was different . . . I cannot go back to an urban life, to an 8 to 5 workday in an office, using public transport everyday . . . but we need to survive in an environment that is not the ideal, so we also need to adapt, [and] sometimes we need to compromise.”

This is a commitment to withdraw from the labour market, to resist capitalist values; it represents the emergence of an ecological habitus and the practice of degrowth at the micro-level. Two years after this first conversation, I talked to CL again. By now she had started her own permaculture community from scratch, but it was not working out very well, so she scrapped the idea of creating a permaculture community. She talked as follows about her experience:

“I was working hard in what I thought would be a permaculture community, building a communitarian house, following permaculture designs, but I was the only one with a permaculture background. The others were not aware of it. I had to convince them of the simplest things related to permaculture, but it did not work. Then I decided to sell my share in the community, and now I’m starting again, taking another course

on permaculture design, and looking for a new plot of land where I can start my own individual homestead . . . ” (CL)

HT, CL, MA and several others I met had entered the permaculture universe and made a decision to move away from their established careers and urban lives. They had confronted a challenging transition period to the accumulation of new forms of capital. The transition period is a time to use one’s accumulated economic capital to pursue new skills and knowledge and to become part of a new community, with ecological values. An example of this transition comes from a couple of PEs who moved away from their professions, one of whom comments on the learning process of building new skills:

“We have totally different backgrounds. I worked in marketing, and my wife was a nurse, [but] we have made a transition to permaculture. It took some years: we moved more and more into permaculture and at the same time moved gradually away from our previous careers. Now we dedicate our time only to permaculture: we produce and sell organic bread and organic cosmetic products, and my wife works in education. We are learning new things every day, it is a matter of trial and error, everything is new, we learn from other PEs, we exchange knowledge all the time. It has been hard sometimes . . . but I never thought of giving up. We’ve re-discussed our path from time to time, like reviewing a path we were going along, whether it was right, whether it needed to change and such. From time to time we do it, but then it’s a function of design, you always re-design, reorganize the house. Permaculture is continuous experimentation . . . ” (MR)

New beginners in permaculture move into a new universe, but to make a living from permaculture without any previous knowledge and experience of how to work with agriculture is a tough endeavour; consequently, they need to acquire new forms of cultural capital with the potential to generate an income from their labour. The wannabe PEs know that their incomes are likely to be reduced by working with permaculture, but permaculture also demands the adoption of a frugal lifestyle and that they detach themselves from their previous middle-class consumer lifestyles. When asked about the main obstacles towards the new lifestyle, HT’s analysis is very illuminating, linking the problems of economic capital with ways of obtaining new cultural capital:

“When I thought I would like to quit my job, I was already married, I already had my first son and I had already incurred debts. I had a twenty-year mortgage, I had a car that I would have to pay for during the next five years, so I thought, I can’t be irresponsible, I have to leave with a reserve before I risk living off the activities I started to develop, but let this reserve guarantee me at least a year, at the very least, without having to take money from anywhere. So, I had to take precautions about that. I was only able to call the boss and announce my resignation when my spreadsheet said: you already have money for one year. This is the main difficulty, which is so difficult that, even if you plan, you end up running out of money because the money runs out in a year and you still haven’t been able to develop the activities you imagined and you still have a rope wrapped around your neck. The difficulty is always the financial issue and the whole turnaround. However much there has been a change in values, philosophical and spiritual, this thing of dealing with money, of abolishing the credit-card paradigm: of using credit, using money that you don’t have sometimes to buy things you don’t need. Abolishing this and starting to live with the money you have on the day, if you don’t have you don’t buy things, this is the main difficulty. It’s a total reprogramming, because we were trained for something else.”

The encounter between the habitus of a middle-class person and the field of permaculture is mostly likely to produce circumstances in which the habitus and the field are “out of sync” with one another. Individuals can choose to go back to their previous fields or to fully enter the field in which their habitus and the field do not complement or reinforce each other. Following the second option involves the pursuit of new cultural and social capital and the emergence of a transformed habitus. A different world view emerges along the way of acquiring different and new types of capital in the field of permaculture and, consequently, individuals begin nurturing an ecological habitus. However, this transfor-

mation takes time, and it is a challenging period in their lives, as they need to reconfigure their set of skills and values and establish a distance from previous social connections. Their skills cannot be used to make a living in the same way as before, as they also need to move away from their previous lives and world views, but they are likely to lose a great amount of cultural capital, and their social capital may become a dead weight, as these social relations have expectations that permaculture entrants are no longer willing to fulfil. The transition from belonging to an urban middle class to a life within permaculture is an uncomfortable choice, immersed in uncertainty and risk. The transition period is the time to reinvent themselves through the pursuit of an ecological habitus to become a member of a permaculture community.

8.2. Permaculture Entrepreneurship and the National Welfare System

The changes in cultural and social capital bring about a transformation of world views, and criticisms of the current social order are articulated. The implications of these changes to one's world view are profound and widespread among PEs; however, families with small children have more difficulties in moving towards what they consider to be a position at the dangerous edge between an ecological life and living as poor families. In many areas they must articulate criticisms of the consumption society but, concerning the education and health of their children, those criticisms have not yet been incorporated into practice.

Brazil is a country of dramatic social inequalities, in which middle-class individuals distinguish themselves from the poor through their higher consumption levels [60], while access to private health and education are central components of middle-class consumption. A middle-class family is likely to have access to private health insurance by paying it directly or through the employer's health insurance of a family member. Health insurance and access to private hospitals are taken-for-granted goods of the middle class, and even the lower-middle class pay for more affordable health insurance.

The public health system is mostly used by the low-income population. The Brazilian public health system is driven by the principles of the so-called Unified Health System (SUS), which has increased access to health care for a significant proportion of the Brazilian population, mostly its low-income citizens. Until recently, several advances were made, with investments in science and technology, primary care, and a substantial decentralization of health care [61]. However, in spite of the clear improvements and heavy investments, the system is inefficient and presents many problems.

When a person decides to move into permaculture, it means moving to a situation of self-employment and, very likely, a lower level of income. The possibility of losing access to private health care could become a central issue for a potential PE. Another important good for the middle class is private schooling for their children: here too, moving into permaculture with a lower income can be problematic for some middle-class individuals:

"When moving into permaculture, private education for the kids and health insurance could not be excluded from our list. We have small children, and to depend on the public health system or public schools is not a good thing" (ME)

The middle class does not entirely trust the public health system, though several PEs have a different attitude, as they have a more complex image about the way health is understood and managed in Brazil. Most permaculture farmers reinterpret the role of private health insurance in their lives, stating that they use the public system combined with alternative medicine. Having a reduced income drives them to be less dependent on private health insurance, and even less reliant on mainstream medicine, so to become dependent on the public health system does not represent a barrier to their moving into permaculture. PEs argue that the social organization of medicine in Brazil needs to be revised, and mainstream medical knowledge is treated with some restraint. Interviewees were fiercely critical of the simple treatment of patients by mainstream medicine:

"When people talk about treatment, it is embedded in their discourse, the idea of individual treatment, so they are more likely to treat the effect in the individual than the cause of

the disease, which is more likely to be caused by the way we live in contemporary society.” (CD)

“We don’t have private health insurance, we’ve never had health insurance, we used the SUS. For some time now, Eve (his wife) needed physical therapy . . . the SUS was creating a lot of difficulties. We tried some alternative ways and it went a little slower, then we paid for some private consultations to be able to solve the problem, and it wasn’t cool, so we decided to pay for health insurance. We’re going to look for other alternatives as well, other medical alternatives for later. We see that mainstream medicine itself does not provide what we want.” (MR)

The transformation towards an ecological habitus challenges the previous ways of understanding several areas of economic and social life, not only how to produce and consume food, but also how health and education are managed in contemporary society. Permaculture practitioners struggle to find new ways to manage their own private health, choosing to be less dependent on private health insurance, paying more attention to prevention, but struggling to keep their children in private schools, which are essentially a core value for the urban middle classes. The education of their children is still an unresolved question for many interviewees, as they consider public schools unable to provide a quality education, and there was no criticism of the social capital acquired in a private school. Permaculture practitioners move away from their social capital and rebuild their cultural capital, but the choices they make for their children are still embedded in a middle-class habitus.

8.3. The New Habitus and Capitalist Relations

PEs make radical criticisms of the negative consequences of the growth economy for the earth and society, not only industrial food production, but also of the organization of urban spaces, the construction of houses, the treatment waste and sewage, and the whole industrial society that they see being imposed on them. The permaculture movement proposes a new way of living on earth. Consequently, the role that a “normal” job should play in their lives is also subject to an intense re-evaluation. Once people start moving into this new universe, the promise that an industrial or service job might be more than just a means to an end and can deeply fulfil the expectations of employees as human beings comes to be highly doubted, and eventually it becomes a worthless cultural conception. It then becomes unbearable to continue as an employee in the “normal” labour market. PEs then try to find ways out of the labour market: *“I was planning my escape for some years”* (HT). However, to step aside from the labour market also means letting go of a rich amount of accumulated capital of different types. On the other hand, once they leave the labour market behind, it is likely to be very difficult to find a way back, this being a permanent turning point. To leave the job market means abandoning most of the social and cultural capital which has been accumulated over the years. The exchange rate for these forms of capital may be very low in the new environment, a representative statement on which was made by a PE:

“I had planned leaving my job for years, but when I did it, I realized that I could do nothing, I had to learn how to survive in this totally new environment. I saved money to pay the bills for a little more than a year, [but] it was not enough. At least two years were needed to learn a new skill and be able to support my family. It’s fine now, but I faced a tough time.” (HT)

The need to acquire new cultural capital is quite clear: taking up permaculture means rejecting a great part of their previous cultural capital. Now, these individuals need to learn how to cultivate, how to build a house, how to avoid producing waste, how to recycle—so many new and different activities and new knowledge to be learnt. This requires a transition period in order to go through almost a metamorphosis of one’s habitus. Another PE (PO) talks about the transformation of the habitus and the search for new solutions to old problems, which implies the search for new types of cultural capital:

“Permaculture is less about technicalities and more about the changing world view, the transformation of values: when you start entering this new universe, you start paying attention to several things that you could not see before. You find new solutions to old problems, but you find problems where there were none. If you are about to build a house, you are not going to buy cement and bricks: you look around, you try to find materials that are already around you and start thinking about the material for each new project, and always try to minimize it. You do not contract people to build your house, you invite friends to help you. They bring knowledge and goodwill, and you can be surprised at how many people show up to help and to work hard for free.” (PO)

When starting to become involved with permaculture, understanding the basic premises that support it and how people live immersed in it, newcomers start thinking critically about the role of work and a career in their lives. Permaculture principles inspire them to question current work values and ways of living in a social order that is driven by capitalistic values. Permaculture promotes a new way of living, and PEs eventually start pursuing change through it. HT explains his personal transformation as a political transformation. He assumes that the labour market imposes levels of specialization that prevent the full development of a variety of skills:

“It is necessary to learn new skills, to search and develop them, in a process of trial and error. In the beginning my drive was that I was quite dissatisfied with many things in my working life . . . I was in pain, not only psychological pain but physical pain as well. From an office in a multinational to agroforestry, that’s a radical political change towards life . . . I became, after all, many things. I had to rescue skills that I had been prevented from developing and that until recently my grandmother had them all. She even had more, which allowed her, for example, to live off of the house. So, there is a guiding movement . . . which is to try to rescue the house as a means of production, to free the house from the reduced condition of a mere centre of consumption.” (HT)

The lack of knowledge and skills has direct implications for levels of income. During the transition period, the PEs experienced growing challenges in realizing their self-imposed choices of moving into permaculture. For instance, many PEs used a lot of time controlling invasive pests in their cultivation, and sometimes they were not able to save their harvest, as they lacked the skills and knowledge to prevent the pests. They could not devise divergent strategies based on their previous work experiences as these were so far apart from their current demands. However, these challenges are not seen as negative, but as part of a learning process in becoming a PE.

The self-imposed confrontation of contrasting world views drives a change towards the ecological habitus, the reproduction of social structures by the agent seems more unlikely, the rejection of previous ways of living more urgent, similar to what Michael Bell calls a “phenomenological break”. The transformation of habitus asks for a profound transformation of the cultural capital carried by those taking up permaculture, and asks for a new place in the division of labour. The new ecological habitus demands that the self be rescued from capitalist relations and drives the search for a reorganization of skills. PEs must avoid the specialized skills that are imposed on them by contemporary society. On the surface it may seem a simpler life, but as the quote above shows, it involves the rescue of older skills and ways of living where the individual needs to be “many things”. It is not a romanticizing of the past: there are many difficulties and barriers to becoming a PE, and their endeavour does not seem to be an easy one. This individual and collective metamorphosis seems to have no way back—their sources and the most value of the previous cultural and social capital are gone. Their previous social and cultural capital are then interpreted as broken chains.

8.4. Building New Social Capital

When adopting permaculture, new entrants are likely to move into a crisis in their lives. What they had been told does not hold true anymore, and they start reevaluating

everything, as their cultural and social capital no longer has much value. In order to overcome this situation, they need to build new social and cultural capital, rejecting their previous working lives and social relations. In order to become a PE, a permaculture entrant needs to reshuffle her life, social connections, skills, job and family relations. The cultural capital needed in this journey is formed through socialization processes with permaculture practitioners, being created and absorbed in repeated interactions with acquaintances and peer groups, as well as through formal education. The creation of social capital among the permaculture community is made through different channels. There are many forms of contact and articulation between those who are interested in permaculture and the current practitioners: social media channels such as YouTube, Facebook, Instagram and WhatsApp are widely used as forums for marketing products and services, as well as for active cooperation and knowledge exchange. Crucial information is distributed through these social media channels. Through these channels, new entrants move away from their previous networks of social relations and invest time in building new social ties to participate in a supportive community that shares their new values and transformed world views. The new social capital has many functions in this community, helping entrants and PEs to reinforce links, and gain and maintain support for their moral and ethical statements and positions. These relationships reinforce their life choices and protect and help them at difficult moments.

Our interviewees claimed that, in moving into permaculture, they faced strong resistance from family members and friends. For instance, the patterns of consumption that correspond to belonging to certain social groups could not be maintained, not only due to lower levels of income, but also to changes in world views. Social ties and forms of consumption are interconnected. Thus, urban middle-class individuals are so convinced of the value of the consumer society that it becomes very difficult for them to accept the possibility of someone voluntarily assuming a frugal lifestyle. As a result, the changes in the world views of PEs become a dividing line. When asked about family and friends, several participants stated that they have difficulties maintaining ties with them:

“There are some who don’t even come here. There are some who think it’s cool, but think it’s cool from the outside: ‘it’s cool, it’s cool to live like this’, but they don’t come to spend any time here, and when they show up, they soon ask, “Where’s the television? Where is this, where is that . . . ” (MR)

“People in my family thought it was a kind of brief trend in living. They did not believe I would change and really engage with permaculture, yet here I am, several years already. Over time I lost contact with some of them. It’s like living in different worlds.” (PT)

Iponema, a Brazilian permaculture association, made a very incisive point about this in their New Year wishes on Facebook:

“For those who are environmentally conscious, respect others, believe in science, wish for the [COVID-19] vaccine and better days, we wish you cheer, health and strength to continue in this New Year! We hope to learn how to take a few more steps in search of a fully sustainable life! This as we understand should be the true new normal!! For the denialists, bolsominions and the like, we wish you luck!”

The right-wing voters of Bolsonaro (called bolsominions, or “Bolsonaro’s minions”) are not welcome among permaculture networks: those who do not respect ecological principles and solidarity should not become a part of the network. Some PEs state that they do not sell their products when they know the buyer is a bolsominion. PEs use different strategies to move away from their previous forms of social capital, creating clear line between “us and them”. When PEs clearly express their aversion to right-wing groups, they demonstrate the distance they keep from the conservative, consumerist society.

Some PEs open their homes to paid visits so that people can get to know about permaculture, gardening, the production of food and sustainable living. Visitors can learn how permaculture homesteaders maintain and use the land, plants and animals, and how to adapt permaculture concepts to urban houses and apartments. Workshops are also held

to teach people how to make ecological products. These activities are often part of the PEs' income, as well as being a way to promote their products. A rich diversity of people attend these workshops. One homestead had received more than a thousand visitors over the past four years, people from all walks of life, genders, economic status, social status and educational background. This is also a practical approach to diffusing information and creating social capital.

Another important event for the permaculture community is the annual convergence, which is intended to reinforce the already existing connections to other PEs and to create new relations, the main objective being to diffuse knowledge and expand the links between the participants. The author participated in one such convergence and noticed how participants mapped the whole community of PEs across the state, how they exchanged information about their new experiences and experiments, discussed attempts with different products, as well as their failures and difficulties, and discussed how to solve problems. The annual convergence facilitates connections to other practitioners and makes it possible for them to embed their experiences much more deeply in the community of PEs. Beyond that, PEs also find support in the moments of difficulty they face when moving into this new universe:

"Permaculture is a new journey for me, I believe for most of us; there is no such a thing as a definitive answer. We are experimenting, so it's important to have peer support, other people like myself, who have experienced the same difficulties, failures and successes, people who get to know all the things that you've been up to and who understand the problems you're facing . . . it is often very exhausting to swim against the tide. We need peers who remind us why you're doing it and who put you back in touch with the drives and reasons that made you start the journey in the first place. It is really rewarding . . . I can see now that I've got a really strong support network. It's turned out to be much more interesting and fulfilling than I thought it would be..." (PW)

The permaculture journey is a transformation of the habitus. Most interviewees talked about the transition period in which they transform their previous forms of capital, when they spend most of the time learning how to survive in this new universe, how to make an income, how to build a house, and how to transition from the old to the new life, make friends and build a supportive community. The transition is a period for transforming the habitus, of starting to see nature from a totally different perspective; it is a period for reflection, of interaction with others who share the same doubts about their relationship with the industrial society and their place within it. The transition is also a period to build new relationships with the permaculture community, strengthening the links, building new friendships, and also moving away from their previous network. The transition is the period when the metamorphosis takes place.

9. Discussion and Conclusions

This article is based on a longitudinal qualitative research project focusing on permaculture as a degrowth practice. From a theoretical standpoint, the literature on degrowth has been criticized for maintaining the gap between normative concepts and the social construction of transitions and sustainable practices. In line with the recommendations of critics concerned with how to overcome this shortcoming, this article has described the problems faced by middle-class individuals when transitioning to become PEs. Our theoretical framework draws on the forms of capital and habitus theory of Pierre Bourdieu, who claimed that the interactions between habitus, cultural capital and the field generate a logic of practice. The logic of practice encapsulates many possible choices, but the choices inscribed in the habitus are limited. The habitus can be seen as containing many matrices which demarcate the range of choices that are available to any individual. Following Bourdieu, we contrast the notion of 'culture' as a map—"the analogy which occurs to the outsider who has to find his way around in a foreign landscape"—with "the practical space of journeys actually made, or rather of journeys actually being made" [30]. We have analysed the changed types of capital of individuals who choose to live sustainably using

permaculture. We have shown that, when deciding to join the permaculture community, actors re-invent their habitus, creating a new world view in order to scrutinize opportunities and constraints and choose the possibilities they want to pursue. Within this new framework, previously inconceivable possibilities are accepted and a new range of choices is constructed, while many previous options become unavailable and even offensive. We have shown that the move from an urban middle-class existence to the life of a sustainable entrepreneur is a long journey of transformation. PEs need to radically adjust their previous cultural maps, thus reimagining the world they want to live in.

Our analytical contribution shows that, when entering into contact with permaculture, individuals become highly uncomfortable with what can be called their own cultural roots and consumption patterns, and struggle to move away from their implicit cultural values. Their habitus develops a lack of integration between their present and past experiences, paving the way for new and unexpected biographies. The discomfort triggered by the encounter with permaculture values drives actors towards an epiphany. Eventually these individuals search to transform their lives, rejecting and struggling to transform their current forms of accumulated capital and, ultimately, their habitus. Newcomers to permaculture face great strains as they move away from their cultural and social capital, which, at that time, were tied to their formal careers in the labour market. Our analysis indicates that their progressive attachment to permaculture happens simultaneously with the transformation of their habitus, which, in general, represents a very difficult and arduous life change because it is seldom supported by the members of the social networks the actors belong to. It is by accumulating new social capital that they can support each other's choices, at the same time as pursuing new forms of cultural capital.

The first-generation PEs in Brazil have reconsidered their previous middle-class economic status and consumption patterns and moved to a level of consumption which is much lower than their previous situations. However, this transformation is not seen as social decline but as a moral choice: they do not see themselves coming back to their middle-class urban lives. They have moved from receiving a secure monthly salary to a situation of variable and uncertain incomes, from the urban to the semi-rural and rural lives, and from being paid employees to becoming entrepreneurs. This is an arduous process of change which emerges in parallel to the gradual burgeoning of internal contradictions. During the transition period new forms of capital are pursued and accumulated. This cultural transformation plays a critical role in shaping their new life aspirations, identities and habitus, which are fundamental to enacting degrowth in practice.

The habitus of PEs switches to a new bodily disposition, which looks and feels quite different. The transformation in habitus is triggered by a situation of conflict with middle-class values and beliefs when pursuing ecologically oriented living. Our analyses indicate that, despite being based on individual discomfort with the current state of society, PEs pursue a transformation based on collectivism—they do not present their transformation as an individual victory. Instead, they pursue and embrace social responsibility as individuals and believe in a bottom-up process of social transformation.

We have seen that the insertion of actors into the practice of permaculture in Brazil opens up possibilities for profound changes in PEs' forms of accumulated capital. They are practicing degrowth by means of a bottom-up re-organization of their social and economic lives. Over time, their habitus, cultural capital, social capital and, consequently, their forms of social valuation are modified. Brazilian permaculturalists are fiercely critical of their previous forms of owned capital, which are understood not as the facilitators of a sustainable life, but as limiting forms of cultural capital. The development of a new habitus and new forms of social and cultural capital become fundamental for actors who practice sustainable ways of living.

Permaculture appears to be one of the inspiring attempts to display key features of sustainable systems, with trajectories towards desirable futures. While it is still an emerging force and at an incipient stage compared to established systems, permaculture can do much to shape the transition towards a sustainable social order. Permaculture

seems to be a utopia of the possible, simultaneously imagining and creating a future, as is clear from the process of learning by trial and error on the part of Brazilian PEs who do not believe in the capacity of the present-day industrial society and new technologies to support the real changes that we all need to pursue when creating a sustainable society. Consequently, the struggle to move away from their previous jobs and industrial lives is evident among all participants, being part of building an ecological habitus. Social boundaries and bonds are re-imagined and remade, sources of previously accumulated cultural and social capital are reconfigured, and the accumulation of new forms of capital is actively pursued. The ecological habitus is painfully constructed by people who need to erase previous forms of accumulated capital and move permanently into permaculture. PEs struggle for recognition that a world that is socially, ecologically and economically desirable must radically produce, organize and distribute capital differently compared to the world we now live in; a transition that asks for a radical transformation of social values. Brazilian actors who seriously immerse themselves into permaculture drastically alter their social relations, their ways of knowing and their ways of living; only then can we talk about a truly ecological habitus. What makes these ecological value systems so interesting for sociologists and other scholars is that they show us that there are other possible futures, other ways of seeing and living well in our contemporary society. This movement is essential to the project of insisting that the kind of common sense that involves our existence, the expectations of consumption and the relationship with capitalism, which powerful institutions want us to take for granted, are currently being (re)evaluated and abandoned by an increasing number of social groups. There are some limitations to this investigation, which investigates a small group of PEs in Brazil, and there is no reliable information concerning the magnitude of the permaculture movement in Brazil, which also focuses on the transition of middle-class individuals into permaculture. Future research should compare the social transformations that are taking place due to permaculture in different countries and among other social groups. Future research could also analyze the connections between the evolution of the permaculture movement and state structures in different contexts, as well as showing how these state structures can support or constrain the emergence of the degrowth movement. It is also important to analyze how changes in the ideologies of ruling parties may affect the development of the agroecology movement. The work of Chappell in his critically important book *Beginning to End Hunger*, about agroecological movements in Belo Horizonte State in Brazil, provides good inspiration for future research on how degrowth might occur.

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